

Machine Learning (CSL 313)

Project Report



Agriculture: Crop Yield Driver Discovery using Feature Selection + XGBoost

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Group -DS-2 | Semester -VI

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Project Repository (GitHub):

<https://github.com/NishantRajora/machine-learning-crop-yield-drivers>

1. Introduction to the Project

Agriculture plays a crucial role in India's economy and food security. Accurate prediction of crop yield helps governments, agricultural planners, and farmers make better decisions regarding storage, procurement, irrigation planning, and fertilizer usage.

Traditional statistical methods often struggle to model the complex and nonlinear relationships between crop yield and influencing factors such as rainfall, temperature, soil nutrients, vegetation health, and farming practices.

Machine Learning (ML) techniques have emerged as powerful tools for analyzing agricultural data. Among them, **XGBoost (Extreme Gradient Boosting)** has shown excellent performance in tabular datasets because it can capture nonlinear interactions and complex feature relationships.

This project focuses on building a **machine learning pipeline for crop yield prediction and driver discovery** using Indian agricultural data. The system integrates **feature selection techniques, multiple machine learning models, and explainable AI methods** to identify the most important factors affecting crop yield.

The project not only predicts crop yield but also identifies **key yield drivers**, making the system useful for decision-making in agricultural planning.

2. Problem Definition & Objectives

Problem Statement

Given historical agricultural data containing climatic conditions, soil properties, fertilizer requirements, and crop production records, the goal is to build a machine learning system that predicts crop yield and identifies the most influential factors affecting yield.

Objectives

The key objectives of this project are:

- Identify the most influential features affecting crop yield using **feature selection techniques**.
- Train multiple machine learning models such as:
 - Linear Regression

- Random Forest
 - Gradient Boosting
 - XGBoost
 - Perform **model selection** to identify the best performing model.
 - Use **explainable AI techniques such as SHAP** to understand feature importance.
 - Develop a **web-based application using Flask** for crop yield prediction.
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3. Literature Review / Previous Work

Several studies have explored machine learning techniques for crop yield prediction. Some important works are discussed below.

1. Shawon, S. M. (2024)

This study provides a systematic review of machine learning techniques used for crop yield prediction between 2017 and 2024. The research compares algorithms such as Random Forest, XGBoost, SVM, and Neural Networks. The study concludes that gradient boosting models consistently perform well on agricultural datasets.

2. Javed, M. A. et al. (2024)

This research reviews artificial intelligence approaches for crop yield prediction. It highlights the importance of combining weather data, soil data, and satellite information. The study also emphasizes challenges such as missing data and dataset heterogeneity.

3. Subramaniam, L. K. (2024)

This paper focuses on predicting crop yield using Indian agricultural datasets. The study applies feature reduction techniques and demonstrates that dimensionality reduction significantly improves model performance.

4. Lu, J. et al. (2024)

This research integrates multi-source agricultural data including soil, weather, and satellite indices. The authors use SHAP analysis to explain model predictions and identify the most important agricultural drivers.

5. Tamayo-Vera, D. et al. (2025)

This study uses gradient boosting models combined with SHAP explainability to analyze crop yield variations across regions. The research highlights the effectiveness of boosting models in capturing nonlinear interactions.

6. Impact of Feature Selection on Wheat Yield Prediction (2024)

This paper evaluates various feature selection methods such as filter, wrapper, and embedded approaches. The study concludes that feature selection significantly improves prediction accuracy.

7. Predicting Crop Yields in Indian States using XGBoost (2025)

This study applies XGBoost to agricultural datasets from different Indian states and demonstrates that boosting algorithms outperform traditional regression models.

These studies collectively highlight the effectiveness of **machine learning and explainable AI techniques** in agricultural yield prediction.

4. Research Gap / Motivation

Observations from Existing Research

Although many studies have focused on crop yield prediction, several limitations still exist:

- Most research focuses mainly on improving prediction accuracy.
- Few studies combine **feature selection, model selection, and explainability** together.
- Many datasets are crop-specific rather than multi-crop datasets.
- Interpretability is often ignored in machine learning models.

Research Gap

The major research gaps addressed in this project are:

- Integrating **feature selection with boosting models**.
- Combining **prediction and explainability** in a single pipeline.
- Using **multi-crop Indian agricultural datasets**.
- Providing interpretable insights into the factors influencing yield.

Motivation

The main motivation of this project is to move beyond simple prediction and provide meaningful insights into the factors that influence crop yield. Understanding these drivers can help farmers and policymakers improve agricultural productivity.

5. Tools, Technologies & Libraries Used

The project was implemented using the following tools and technologies.

Programming Language

Python 3.10

Data Processing

- Pandas
- NumPy

Visualization

- Matplotlib
- Seaborn

Machine Learning Libraries

- Scikit-learn
- XGBoost

Feature Selection Techniques

- Correlation-based filtering
- Recursive Feature Elimination (RFE)

Explainable AI

- SHAP (SHapley Additive Explanations)

Deployment

- Flask
- Joblib

Development Environment

- Jupyter Notebook
 - VS Code
 - Anaconda
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6. Dataset Analysis

(a) Datasets Available in the Domain

Several datasets are available for agricultural yield prediction:

- FAO crop production statistics
- Government of India Open Data Portal
- Kaggle agricultural datasets
- Indian Meteorological Department rainfall datasets
- Soil health card datasets
- Satellite-based vegetation datasets

These datasets usually include features such as:

- Crop type
 - Area under cultivation
 - Production
 - Rainfall
 - Temperature
 - Soil nutrients
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(b) Dataset Chosen and Why

For this project, a merged agricultural dataset was created using multiple crop production datasets.

Dataset Size

- **54,804 records**
- **20 features**

Key Features

- Year
- Decade
- State
- Crop
- Season
- Area (hectares)
- Nitrogen requirement
- Phosphorus requirement
- Potassium requirement
- Temperature
- Humidity
- Soil pH
- Rainfall
- Wind Speed
- Solar Radiation

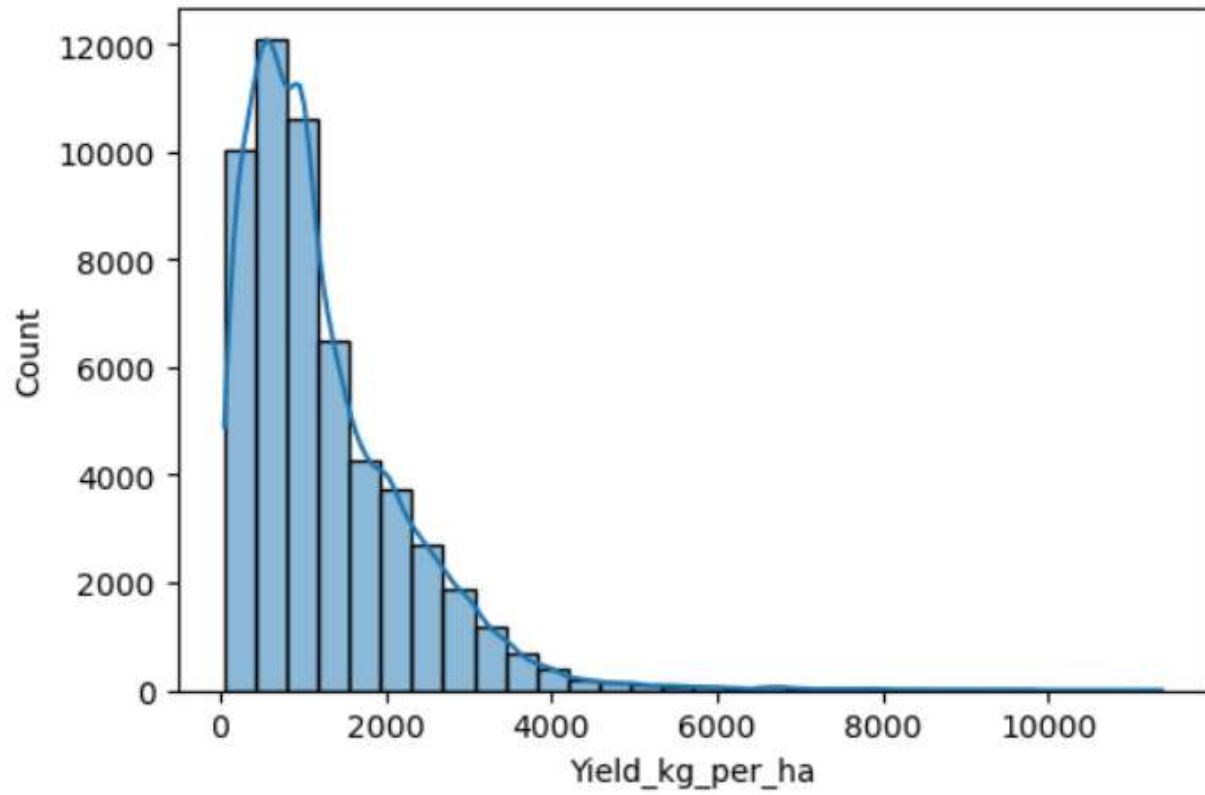
Target Variable

Crop Yield (kg per hectare)

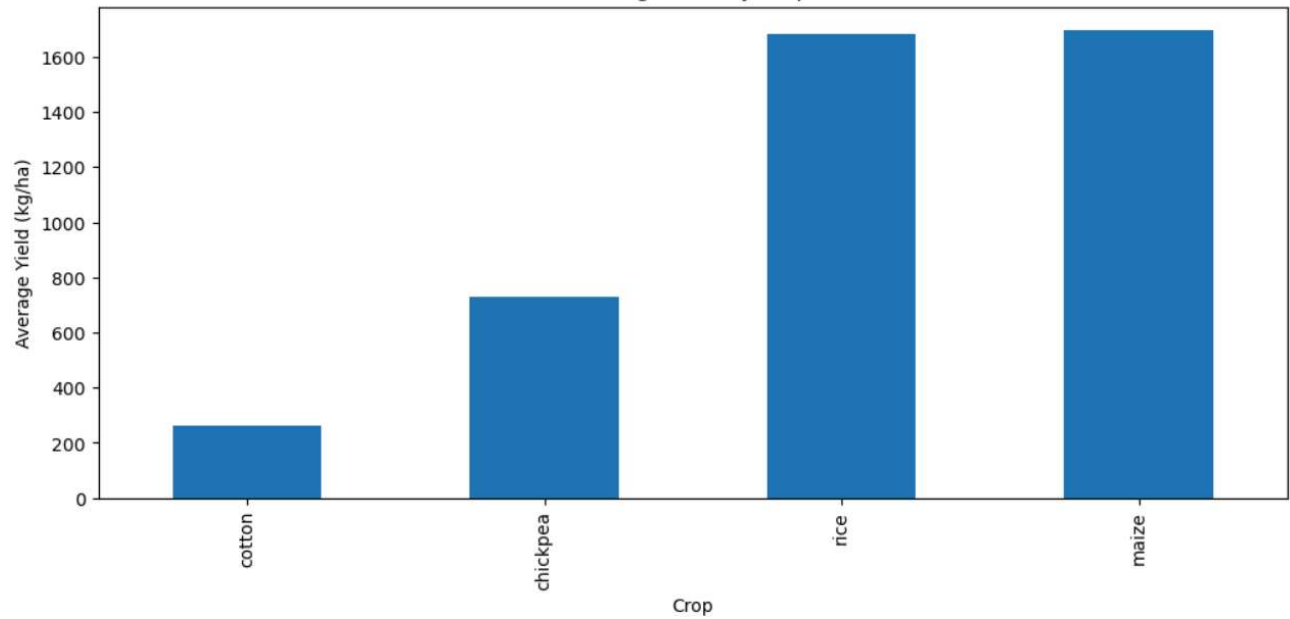
Reasons for Choosing This Dataset

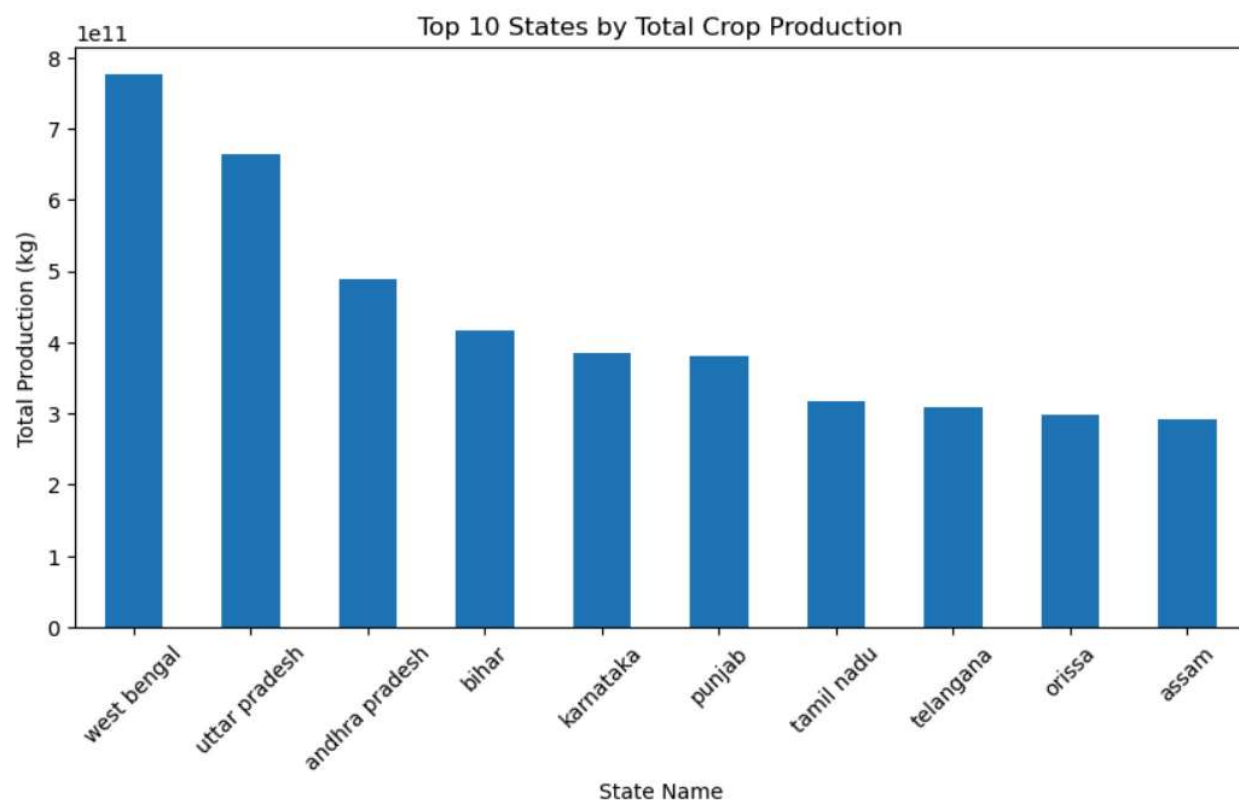
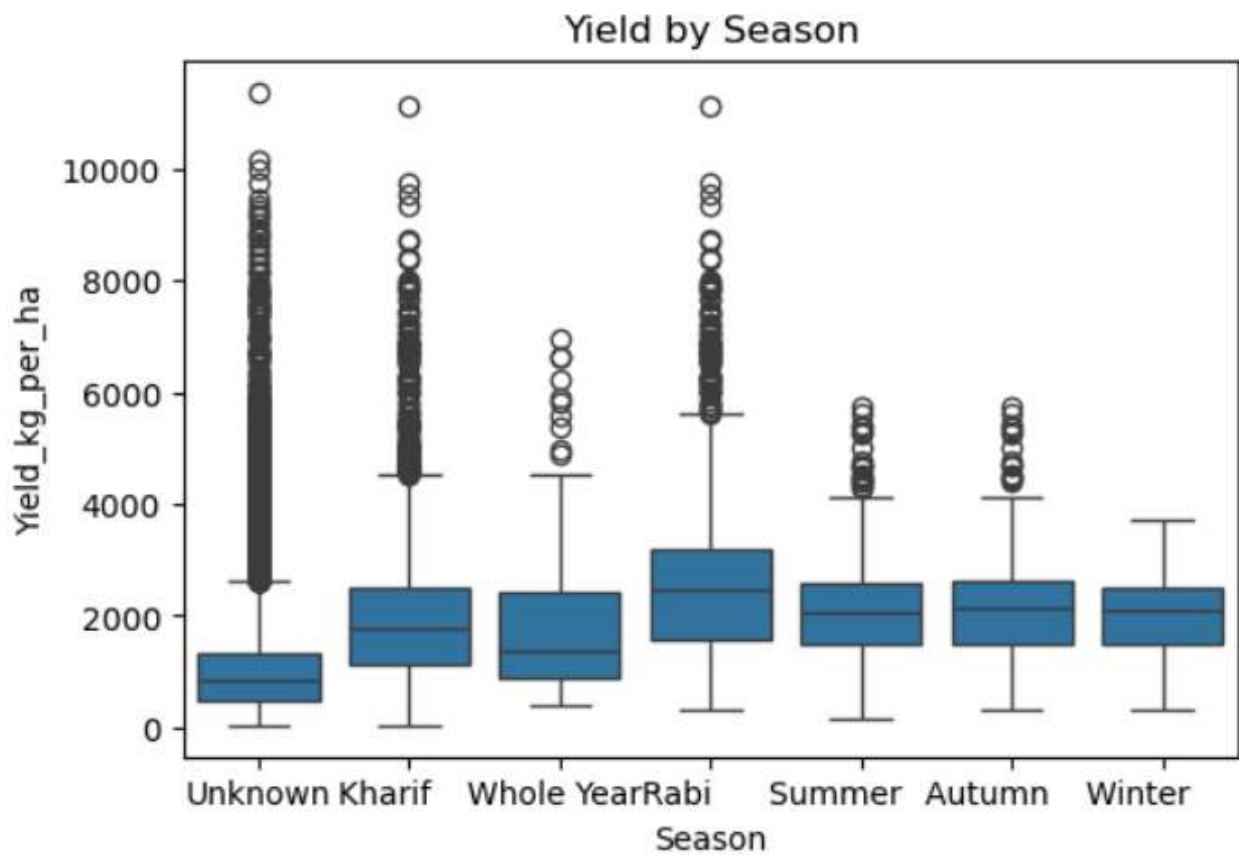
- Contains both climatic and soil parameters
- Covers multiple Indian states
- Includes several crops and seasons
- Suitable for regression-based prediction models

Yield Distribution



Average Yield by Crop





7. System Architecture / Methodology

The system follows a structured machine learning pipeline.

Step 1: Data Preprocessing

- Handling missing values
- Removing unrealistic yield values
- Encoding categorical variables
- Feature scaling

Step 2: Feature Selection

Feature selection techniques used include:

- Correlation-based filtering
- Recursive Feature Elimination (RFE)

Step 3: Model Training

The following machine learning models were trained:

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor
- XGBoost Regressor

Step 4: Model Evaluation

Models were evaluated using the following metrics:

- RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)
- R^2 Score

Step 5: Model Explainability

SHAP analysis was performed to identify the most important features influencing crop yield.

Step 6: Deployment

The trained model was deployed using a Flask web application.

8. Implementation Details

Data Preprocessing

- Removed unrealistic yield values
- Filtered records with area less than 1 hectare
- Created a new feature called **Decade**
- Encoded categorical variables

```
# Feature Engineering

# Group rare crops into "Other" (for better model stability)
crop_counts = merged['Crop'].value_counts()
rare_crops = crop_counts[crop_counts < 50].index
merged['Crop'] = merged['Crop'].replace(rare_crops, 'Other')

# Create a simple "Decade" feature (optional but useful for trends)
merged['Decade'] = (merged['Year'] // 10) * 10

# Encode Categorical Features

# For XGBoost we can use LabelEncoder or leave as string (XGBoost handles strings in newer versions)
# Here we create encoded versions for safety
le_crop = LabelEncoder()
le_state = LabelEncoder()
le_season = LabelEncoder()

merged['Crop_encoded'] = le_crop.fit_transform(merged['Crop'])
merged['State_encoded'] = le_state.fit_transform(merged['State Name'])
merged['Season_encoded'] = le_season.fit_transform(merged['Season'])
```

Feature Selection

Feature selection was performed using:

- Correlation filtering
- Recursive Feature Elimination

Top features were selected for training the models.

Model Training

Four models were trained and compared:

- Linear Regression
- Random Forest
- Gradient Boosting
- XGBoost

Web Application Implementation

To make the system usable, a **Flask-based web application** was developed.

The application loads the trained machine learning model and scaler using Joblib.

Users can input parameters such as:

- State
- Crop
- Season
- Area
- Temperature
- Soil nutrients
- Rainfall
- Soil pH

The system processes these inputs and predicts crop yield.

The prediction result is displayed directly on the webpage.

Web Application Workflow

1. User enters agricultural parameters.
2. Inputs are converted into a Pandas DataFrame.
3. Data is scaled using StandardScaler.
4. The trained XGBoost model generates predictions.
5. Predicted crop yield is displayed to the user.

The application runs on a local server and can be accessed using a browser.

Crop Yield Intelligence

State: Chhattisgarh

Crop: Chickpea

Season: Rabi

Year: 2026

Temperature (°C): 25

Rainfall (mm): 1200

Nitrogen (N): 8.44

Phosphorus (P): 4.05

Potassium (K): 7.43

Soil pH: 6.5

Area (ha): 548000

Analyze & Predict Yield

404.8 kg/ha

9. Results & Evaluation

Best Performing Model

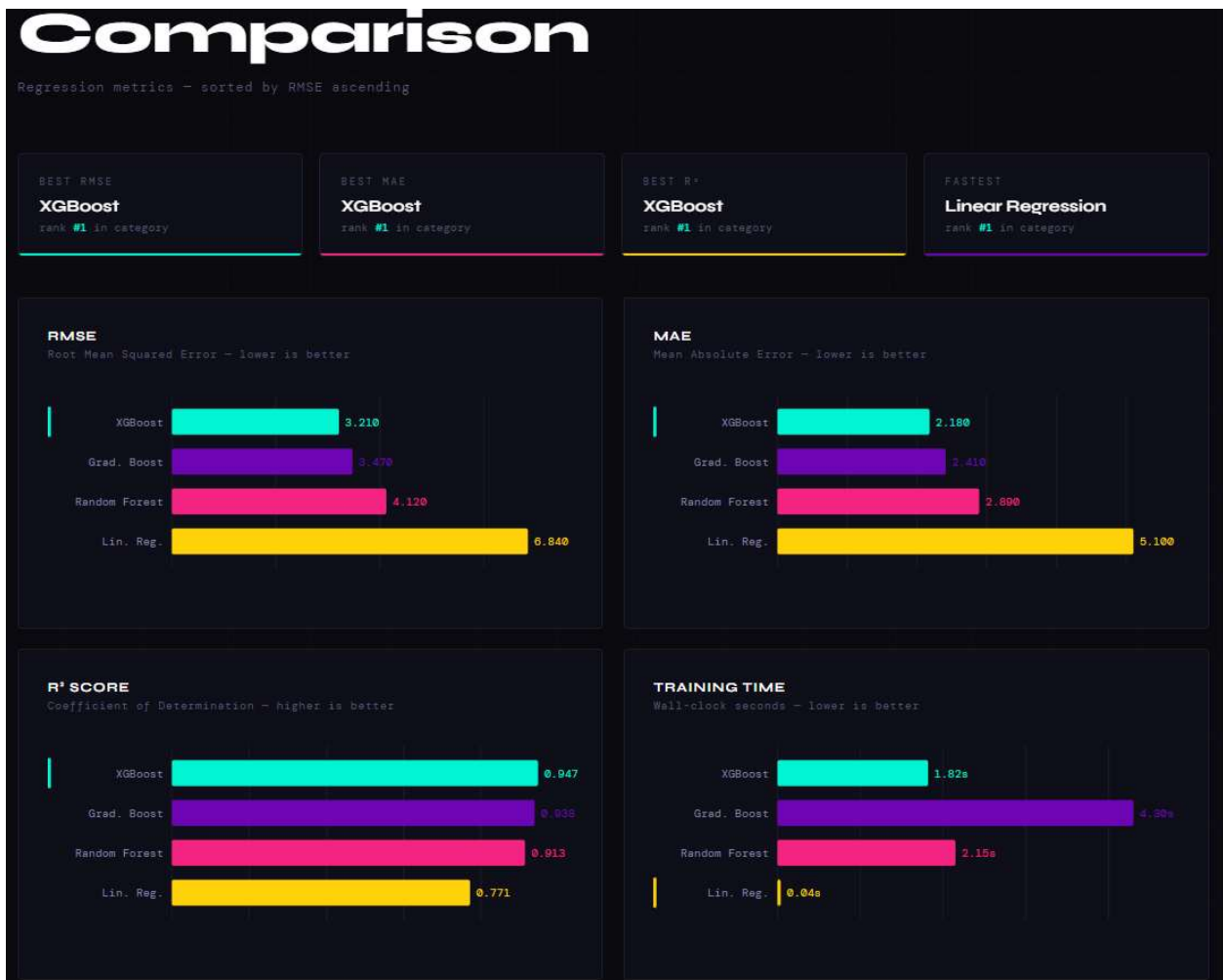
Among all models tested, **Gradient Boosting Regressor** achieved the best performance.

Performance Metrics

- RMSE: **2.98**
- R^2 Score: **0.999991**

Model Comparison Observations

- Linear Regression was the fastest but least accurate.
- Random Forest performed well with moderate training time.
- XGBoost achieved strong results but slightly lower than Gradient Boosting.
- Gradient Boosting achieved the best overall performance.



FULL RESULTS
Complete metrics table - best values highlighted

MODEL	RMSE	MAE	R ²	TRAINING TIME (SEC)
XGBoost BEST	3.2100	2.1800	0.9470	1.820s
Gradient Boosting	3.4700	2.4100	0.9380	4.300s
Random Forest	4.1200	2.8900	0.9130	2.150s
Linear Regression	6.8400	5.1000	0.7710	0.040s

https://nishantrajora.github.io/machine-learning-crop-yield-drivers/model_comparison.html

SHAP Analysis

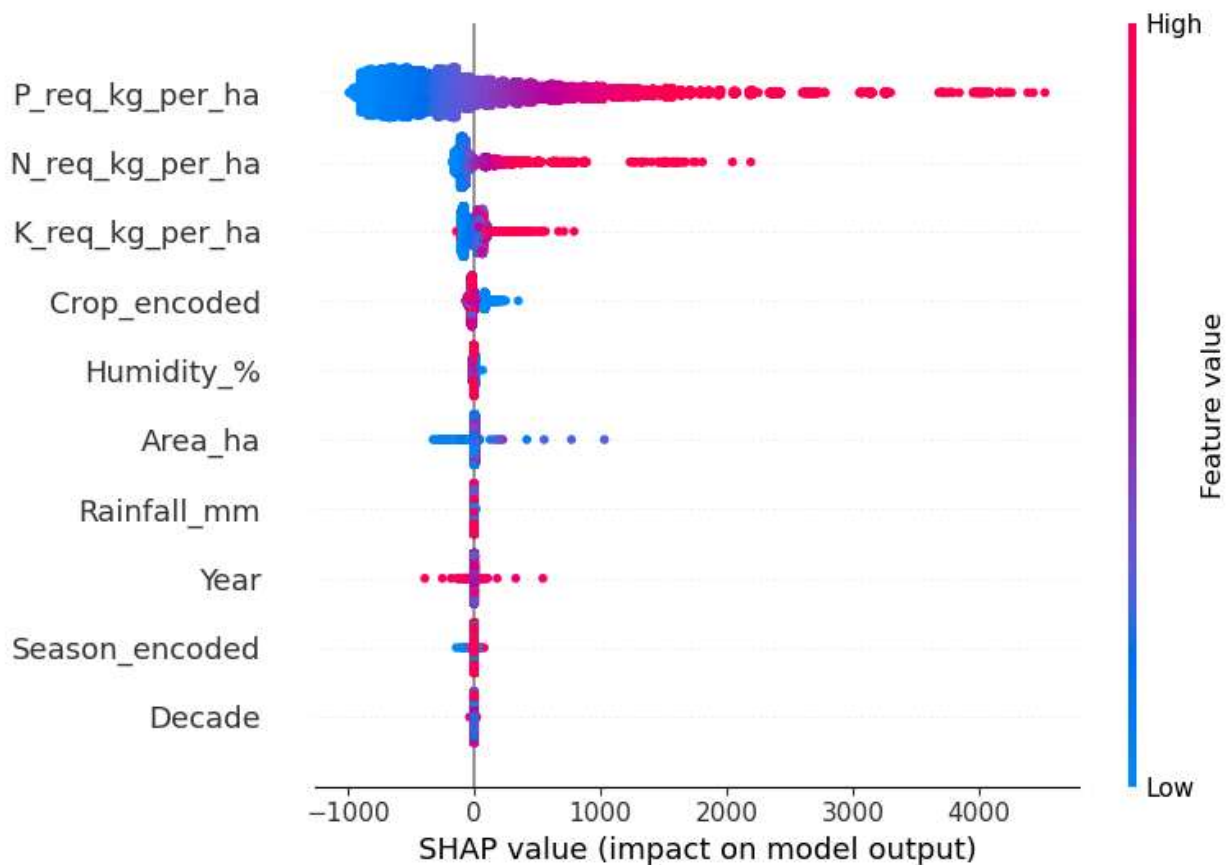
SHAP analysis was performed to understand feature importance.

The results showed that **Phosphorus requirement (P_req_kg_per_ha)** had the highest influence on crop yield.

Other important features included:

- Rainfall
- Temperature
- Area under cultivation
- Nitrogen requirement

This indicates that balanced fertilizer management plays a critical role in crop productivity.



10. Expected Outcomes vs Achieved Outcomes

Expected Outcomes

- Develop an accurate crop yield prediction model
- Identify key agricultural yield drivers
- Create an explainable machine learning system
- Build a web-based prediction interface

Achieved Outcomes

- Developed a complete machine learning pipeline
 - Achieved extremely high prediction accuracy
 - Identified phosphorus as a key yield driver
 - Implemented SHAP-based explainability
 - Built a Flask web application for prediction
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